

Food Waste Reduction Platform: Robust System Design and Project Management

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I. RISK MANAGEMENT PLAN

The Risk Management Plan follows a structured, iterative methodology aligned with PMBOK and ISO 31000 standards, ensuring systematic identification, analysis, response planning, and continuous monitoring throughout the project lifecycle. This plan is fully traceable to the 17 measurable design requirements and the 10 critical risks/sensitivities documented in Workshop 1 and explicitly addressed by the architectural patterns of Workshop 2.

1) *Risk Management Approach*: The approach consists of four integrated stages:

- **Risk Identification**: Derived directly from the critical sensitivities, emergent behaviors, and chaotic dynamics identified in Workshop 1 systems analysis.
- **Risk Analysis**: Qualitative and quantitative evaluation of probability and impact.
- **Risk Response Planning**: Definition of proactive mitigation strategies and contingency actions.
- **Risk Monitoring and Control**: Real-time tracking through KPIs, dashboards, and automated alerts embedded in the Analytics Layer (Workshop 2).

A hybrid strategy combining proactive architectural controls (Workshop 2) and reactive contingency measures was adopted, which is essential for a real-time, high-concurrency platform operating under strict geospatial and temporal constraints.

2) *Risk Register*: Table II presents the ten critical risks identified in Workshop 1, their assigned owners (per the RACI matrix), mitigation strategies implemented in Workshop 2, contingency plans, and monitoring metrics. See Table I

3) *Risk Matrix*: The probability-impact relationship of the ten risks is summarized below (updated after Workshop 3 implementation and testing). See Table II

4) *Monitoring and Control Strategy*: Continuous risk control is embedded in the platform's Analytics Layer (Workshop 2):

- Real-time dashboards visualizing KPIs and risk indicators.
- Centralized logging of all system events.
- Automated alerts triggered when predefined thresholds are exceeded.

Associated monitoring KPIs:

- No-show rate $\leq 15\%$
- Notification delivery time < 5 seconds
- Matching latency < 2 seconds

- Unclaimed surplus rate $< 20\%$

- System uptime $\geq 99\%$

5) *Risk Response Strategy*: The project applies standard response strategies:

- **Avoidance**: Critical technical risks (e.g., R1 – race conditions) via architectural serialization.
- **Mitigation**: Most operational risks (R2–R9) through Workshop 2 design patterns.
- **Acceptance**: Behavioral risks that cannot be fully eliminated (R5, partially R6), managed via monitoring.

6) *Systemic Risk Perspective*: The identified risks form a dynamic system with feedback loops (no-show cascades, notification-driven demand amplification, and behavioral participation cycles). By integrating risk controls directly into the four-layer architecture (Workshop 2) and maintaining closed feedback loops (daily stand-ups, post-experiment retrospectives, and GitHub issue tracking), the project achieves high resilience and ensures long-term sustainability of the Food Waste Reduction Platform.

REFERENCES

- [1] Workshop 1 – Food Waste Reduction Mobile Application, A Rigorous System Analysis, Universidad Distrital Francisco José de Caldas, 2026.
- [2] Workshop 2 – System Design Document, Universidad Distrital Francisco José de Caldas, 2026.
- [3] IEEE Standards Association, “IEEE 730-2014 - IEEE Standard for Software Quality Assurance Processes,” 2014.

Table I: Critical Risk Register (traceable to Workshop 1)

ID	Risk Description (Workshop 1)	Owner	Mitigation (Workshop 2)	Strategy	Contingency Plan	Monitoring Metrics
R1	High-demand surplus events causing race conditions	Backend Lead	Serialized priority-queue + DB locking		Fair lottery allocation	contention_rate
R2	High concurrency degrading performance	Backend / DevOps	Redis caching + rate limiting		Graceful degradation	concurrent_users, response_time
R3	Geolocation service failure (fragile GPS input)	Backend / Infra	Redundant services + client-side validation		Manual location fallback	geo_failures_per_hour
R4	No-show after acceptance	Backend / Product	Reliability scoring + automatic timeout		Automatic reassignment	no_show_rate
R5	Informal external coordination (WhatsApp bypass)	Product Owner	Anomaly detection + audit logging		Ranking penalties	anomaly_rate
R6	Supplier disengagement due to low demand	Project Manager	Engagement metrics + feedback loops		Targeted push notifications	supplier_activity_trend
R7	Inequitable distribution due to proximity bias	Backend / Data	Fairness-weighted matching algorithm		Dynamic weight adjustment	equity_index
R8	Cascade failures from no-show chains	Backend	Limited reassignment cycles (max 3)		Stop and escalate	reassignment_chain_length
R9	Demand amplification from notifications	Backend / Mobile	Staggered tiered notifications		Batch-size limiting	notification_spike
R10	Negative feedback loop from poor metrics	Analytics / PM	Contextualized analytics dashboard		Personalized performance reports	publication_rate

Table II: Risk Probability vs Impact Matrix (post-Workshop 3)

Impact \ Probability	Low	Medium	High
High	R5, R7	R6, R10	R1, R2, R3, R4, R8, R9
Medium	–	R5	R6
Low	–	–	–